PACMAN Programmable Logic Firmware 3.6

Global, TX, RX and Timing units

Draft maintained by the PACMAN agents (augie); reference for the TRM

2026-10-17

1 Purpose and scope

This note describes the four main units of the PACMAN programmable logic (PL) firmware, version 3.6 (candidate release), as built for PACMAN 1v5:

- **Global**: board-wide settings (tile power and enables, LEDs), status, and version information.
- **TX**: transmission to the tiles over 40 UART channels (POSI), from DMA.
- **RX**: reception from the tiles over 40 UART channels (PISO), to DMA.
- **Timing** (ATC, ASIC timing and control, in the firmware sources): the ASIC clock, the bit timing of the UARTs, the timestamp, and the control signals G and H sent to the tiles.

The ADC unit is not covered. The register addresses are those of `register_map.vhd`, which is the source of truth; the behavior described is that of the VHDL in `ghdl/` of the `pacman-firmware` repository. Every timing diagram is the output of a GHDL simulation of the unmodified RTL, made by `make waveforms` in `ghdl/agents/`, so the figures can be regenerated whenever the firmware changes.

2 Common structure

2.1 Clocks and data paths

The PL runs from two 100 MHz clocks from the processor (PS). The first clocks the AXI interfaces, the DMA engine, the register bus bridge, and the Global and ADC units. The second clocks the TX, RX and Timing units, which handle all signals to and from the tiles. The two domains meet in three places, each designed for clock domain crossing: the register bus hub, the AXI stream FIFO between the RX unit and the DMA engine, and the AXI stream FIFO between the DMA engine and the TX unit (both FIFOs hold 64 words of 64 bits).

Data to be transmitted is written by the PS to memory and streamed by DMA (MM2S) to the TX unit. Received data is streamed by the RX unit to DMA (S2MM), which writes it to memory for the PS. The Timing unit provides the TX and RX units with the bit timing of the UARTs (the BAUD strobe), and the RX unit with the timestamp and the marker pulses. The TX outputs (POSI) are also wired back to the RX unit, for a loopback test without any external connection.

As in the TRM, transfers use the valid and ready handshake of the AXI protocol: data moves from producer to consumer on a *beat*, a clock cycle in which the producer asserts valid and the consumer asserts ready. The same handshake is used inside the units, between the buffers and the 40 UART channels.

2.2 Register bus

Registers are reached over a simple custom register bus (REGBUS). A bridge (`axil_to_regbus.vhd`) turns AXI-Lite transactions from the PS into REGBUS requests, and a hub (`regbus_hub.vhd`) dis-

tributes them to the units: directly to the Global and ADC units, and across the clock domain crossing (`regbus_cdc.vhd`) to the TX, RX and Timing units.

A write request is one clock cycle with the write update bit high, the 16-bit write address, and the 32-bit write data. The unit that owns the address sets the register and raises the write acknowledge for one clock cycle, on the next clock cycle. A read request is one clock cycle with the read update bit high and the 16-bit read address; the unit answers with the 32-bit read data and the read acknowledge. The Global and Timing units answer reads on the next clock cycle; the TX and RX units answer reads one clock cycle later (two clock cycles after the request), because of an additional register stage on their read data. A write and a read of the Global unit are shown in Fig. 1.

Only defined registers are acknowledged. A request to an undefined address, or a write to a read-only register, changes nothing and is never acknowledged. The bridge waits at most 256 clock cycles for an acknowledge; it then completes the AXI transaction anyway, so that the PS never hangs, and a read returns `0xEEEEEEEE`.

Table 1: Register address scopes. Addresses are 16 bits; the most significant four bits select the unit.

Unit	Scope bits	Addresses	Register field
TX	`00XX`	`0x0000`–`0x3FFF`	channel (6 bits) and register (8 bits)
RX	`01XX`	`0x4000`–`0x7FFF`	channel (6 bits) and register (8 bits)
(reserved)	`1000`–`1100`	`0x8000`–`0xCFFF`	
ADC	`1101`	`0xD000`–`0xDFFF`	register (12 bits)
Timing	`1110`	`0xE000`–`0xEFFF`	register (12 bits)
Global	`1111`	`0xF000`–`0xFFFF`	register (12 bits)

The 16-bit address space is divided by scope as shown in Table 1. The TX and RX units hold registers for each of the 40 UART channels, and their addresses are read as

$$0\mathrm{b}\, ss\, cccccc\, rrrrrrrr,$$

with the scope ss (`00` for TX, `01` for RX), the channel $C = 0\mathrm{b}cccccc$, and the register $R = 0\mathrm{b}rrrrrrrr$. Channels `0x00` to `0x27` are the UARTs 0 to 39. Two further channels are notional: *broadcast* (`0x3B`), a write to which sets that register for all 40 UARTs at once (it cannot be read), and *common* (`0x3F`), which holds the registers of the unit as a whole. For example, the configuration register (`0x04`) of RX UART 5 is at `0x4504`, the same register of all RX UARTs is written at `0x7B04`, and the RX buffer configuration (`0xB4`, common) is at `0x7FB4`. In the TX unit, the same configuration registers are at `0x0504` and `0x3B04`, and the TX buffer status (`0xB0`, common) is at `0x3FB0`.

Two conventions recur in the TX and RX units. The *counters* are 16 bits wide and saturate at `0xFFFF`; they are cleared at reset and by a write (of any value) to the unit's ZERO_CNTS register. The *look* registers show the 64-bit word of one UART, chosen by writing its number to LOOK_SELECT: the least significant 32 bits are read at LOOK_LOWER and the most significant at LOOK_UPPER.

3 Global unit

The Global unit handles simple board-wide tasks, through the registers of Table 2.

Powering a tile requires enabling the tile power subsystem with TILE_POWER_ENABLE (called ANALOG_POWER_ENABLE in the schematics) as well as the tile itself, with its bit of TILE_ENABLES. The other bits of ENABLES, and bits 31–2 of LEDS, are stored and read back, but drive nothing. Each of the two PL LEDs is simply on or off, as set by its bit of LEDS; the two

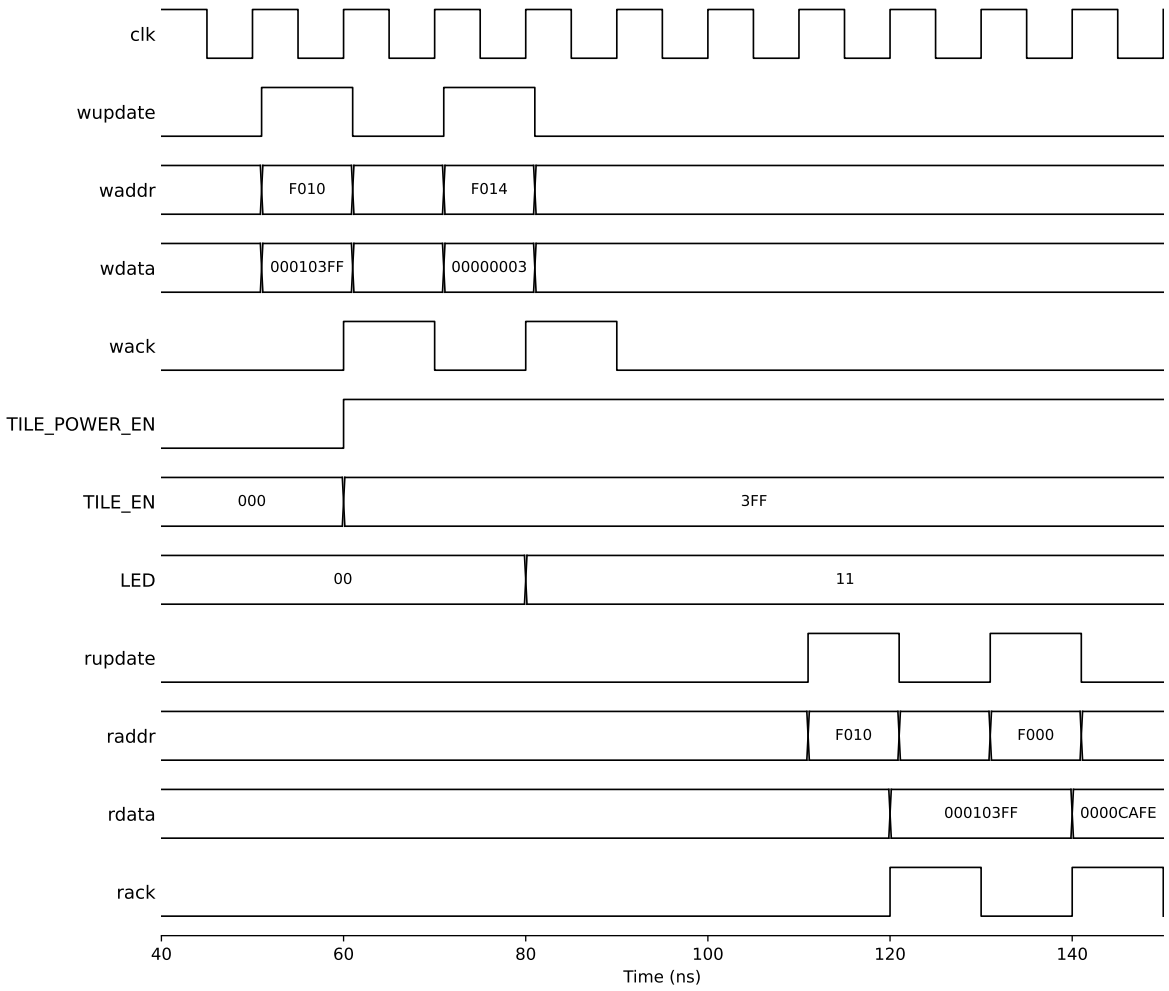


Figure 1: The Global unit on the register bus (GHDL simulation of `global_unit`). Two writes, ENABLES (`0xF010`) with tile power and all ten tiles, and LEDS (`0xF014`) with both LEDs on, each acknowledged on the next clock cycle, at which time the outputs change. Then reads of ENABLES and STATUS, each answered on the next clock cycle.

Table 2: Registers of the Global unit.

Register	Address	Access	Contents
STATUS	0xF000	RO	fixed pattern 0x0000CAFE
ENABLES	0xF010	RW	tile enables and tile power (Table 3)
LEDS	0xF014	RW	the two LEDs driven by the PL (Table 3)
SCRATCH_A	0xF020	RW	no effect: for testing the register path
SCRATCH_B	0xF024	RW	no effect: for testing the register path
FIRMWARE_MAJOR	0xFF10	RO	firmware version: 3
FIRMWARE_MINOR	0xFF14	RO	firmware version: 6
FIRMWARE_PATCH	0xFF18	RO	firmware version: 0
HARDWARE_MAJOR	0xFF20	RO	hardware version the firmware was built for: 1
HARDWARE_MINOR	0xFF24	RO	hardware version: 5
HARDWARE_PATCH	0xFF28	RO	hardware version: 0
SYNTHESIS_DATE	0xFF30	RO	time of the build, in seconds since 1970 (Unix time)
GIT_HASH_UPPER	0xFF40	RO	short git hash of the build, ASCII, characters 1–4
GIT_HASH_LOWER	0xFF44	RO	short git hash, ASCII, characters 5–7 and a space
BUILD_FLAGS	0xFF48	RO	bit 0 DIRTY, bit 1 OFFICIAL_RELEASE
VIVADO_MAJOR	0xFF50	RO	Vivado version of the build (e.g. 2023)
VIVADO_MINOR	0xFF54	RO	Vivado version of the build (e.g. 2)

Table 3: Bit fields of the Global registers.

Register	Field	Bits	Mask
ENABLES	TILE_ENABLES	9–0	0x000003FF
	TILE_POWER_ENABLE	16	0x00010000
LEDS	LED_3	0	0x00000001
	LED_4	1	0x00000002

other LEDs of the PACMAN are driven by the PS. Voltages, currents and the on-board multiplexer are controlled over I2C by the PS, and are not part of the PL.

The version registers identify the firmware without reference to any file. The firmware and hardware versions are constants of the source (`firmware_version.vhd`, `hardware_version.vhd`). The synthesis date, git hash, build flags and Vivado version are written at build time. DIRTY is set when the Vivado sources (`vivado/`) had uncommitted changes at the time of the build; OFFICIAL_RELEASE is set when the build was made from a commit marked as an official release (with the `BUILD_RELEASE` file). In a GHDL simulation these build-time registers all read zero.

As in Fig. 1, the Global unit acknowledges reads and writes on the next clock cycle, and a register written is visible at its output from that same clock cycle.

4 TX unit

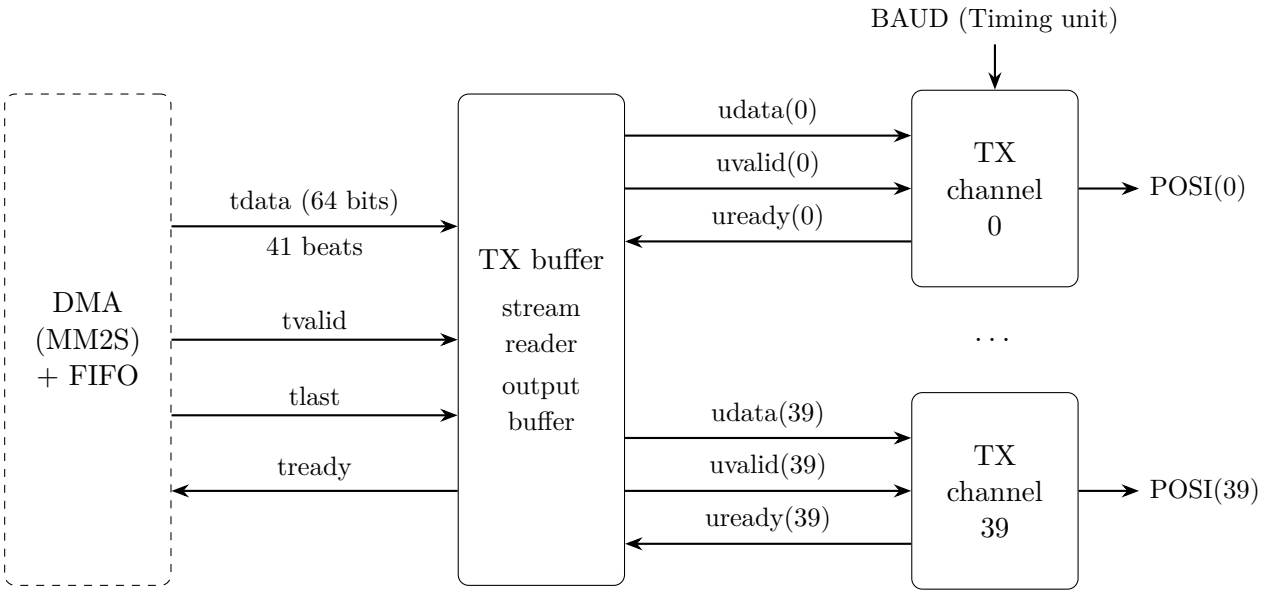


Figure 2: Data path of the TX unit.

The TX unit transmits data from the PS to the tiles over 40 UART channels. Its data path is shown in Fig. 2. The PS fills a DMA packet in memory; the DMA engine streams it to the TX buffer (through a FIFO that crosses into the TX unit's clock domain); the TX buffer hands each UART its 64-bit word; and 40 TX channels, one per UART, transmit their words in parallel on the POSI lines.

4.1 DMA packet

A DMA packet is always 41 beats of 64 bits (328 bytes), as in Table 4: a mask with one bit per UART, then one 64-bit word per UART, in UART order. Only the UARTs selected by the mask transmit; the words of the others are ignored. The packet is the same size however many UARTs are selected. The length is not checked by the firmware: a packet that ends early (TLAST before beat 40) leaves the missing words as they were in the previous packet.

4.2 TX buffer

The TX buffer has two stages. The first, the stream reader, collects the 41 beats of a packet. Once a complete packet has arrived (TLAST), it deasserts tready, and the packet is copied into the second stage,

Table 4: The TX DMA packet: 41 beats of 64 bits, TLAST on the last.

Beat	Contents (64 bits)	Description
0	`0x000000MMMMMMMMMM`	M: UART mask, bit i for UART i
1	`0xDDDDDDDDDDDDDDDD`	D: the word for UART 0
2	`0xDDDDDDDDDDDDDDDD`	the word for UART 1
	...	
40	`0xDDDDDDDDDDDDDDDD`	the word for UART 39

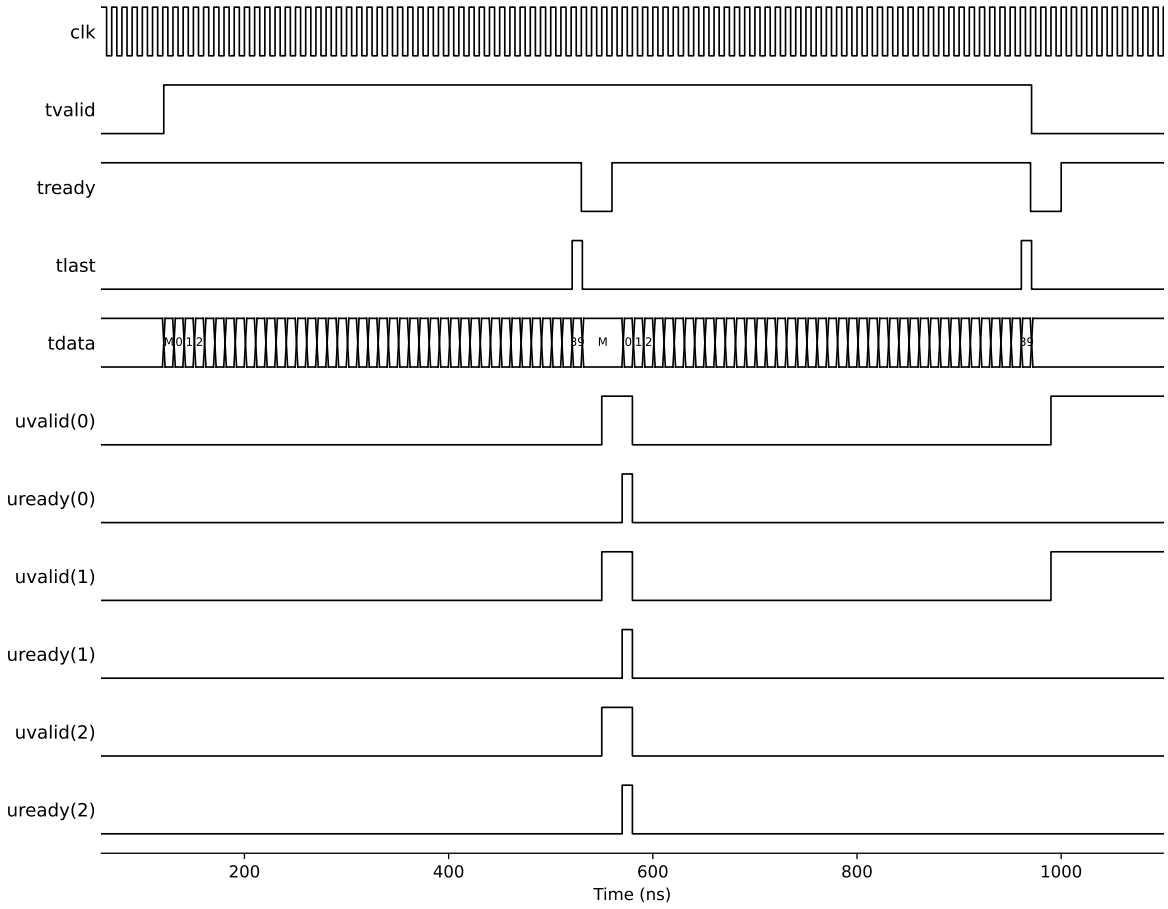


Figure 3: The TX buffer receiving two DMA packets back to back (GHDL simulation of `tx_unit`). Packet 1 selects UARTs 0–2, packet 2 UARTs 0 and 1. On tdata, M is the mask beat and the numbers are the UART of each word (labels shown for UARTs 0–2 and 39 only). The TX buffer drops tready for three clock cycles after each packet, while the packet moves to the output buffer. The channels of packet 1 take their words at once (uready); those of packet 2 are still transmitting packet 1, so the words of packet 2 wait in the output buffer (uvalid stays high, beyond the end of the figure).

the output buffer: one 64-bit word per UART, with a valid bit (uvalid) set for each UART of the mask. The stream reader is then released and asserts tready again, ready for the next packet, while the UARTs transmit. The whole hand-over takes three clock cycles (Fig. 3).

Each TX channel takes its word when it is ready to start a new transmission: it asserts uready for one clock cycle, which clears that UART's uvalid on the next clock cycle. The output buffer accepts the next packet only when every uvalid bit is clear, that is, when every selected UART has taken its word. With the UART shift registers, the output buffer and the stream reader, up to three packets are in the TX unit at once: one being transmitted, one waiting in the output buffer, and one waiting in the stream reader, as in Fig. 3.

The UART transmission is by far the slowest step. At the default bit rate of 10 Mbit/s, a 66-bit frame (a start bit, 64 data bits and a stop bit) takes 6.6 μs, or 660 clock cycles; a DMA packet takes 41 clock cycles. As long as DMA keeps the stream reader full, the TX channels transmit back to back, separated only by their configured delay.

4.3 TX channel

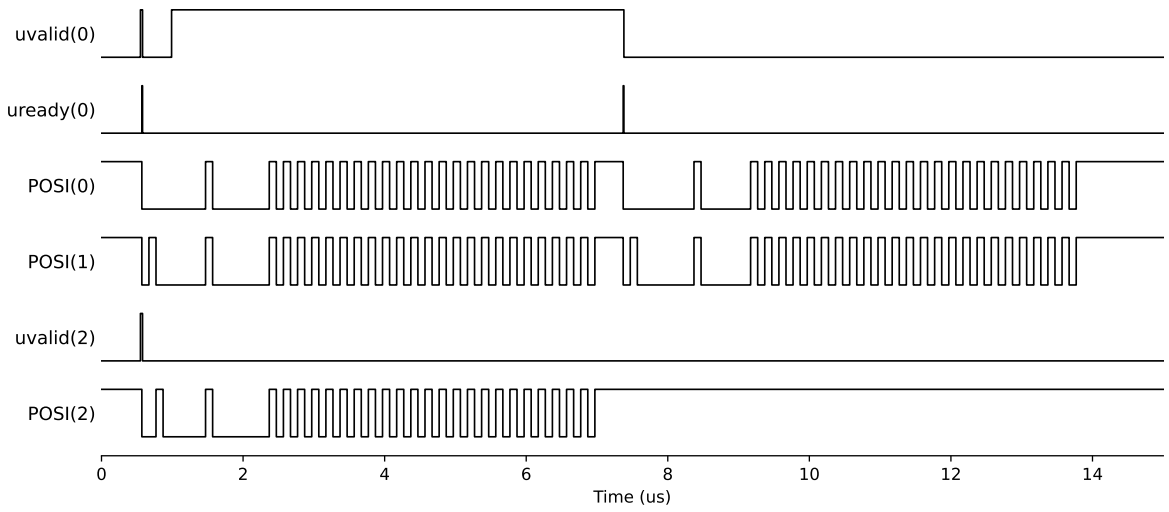


Figure 4: The UART frames of the two packets of Fig. 3 (GHDL simulation of `tx_unit`; all UARTs configured `0x00020006`: PHASE 6, DELAY 2). Each word is `0xAAAAAAAAAAAA` above a 16-bit identifier (`0x0n0u`: packet n, UART u), sent least significant bit first, so each frame shows a start bit, 16 identifier bits, then 48 alternating bits, and a stop bit. UARTs 0 and 1 take the second word (uready) after the stop bit and the 2-bit delay.

Each TX channel is a UART transmitter with a 64-bit shift register. Its bit timing comes from the BAUD strobe of the Timing unit, one clock cycle at the start of each bit period (every 10 clock cycles, or 10 Mbit/s, by default). The channel acts PHASE clock cycles after each BAUD strobe; PHASE sets the position of the bit edges within the bit period, and is the same for all channels unless configured otherwise.

While idle, the line is high. At the first bit boundary at which valid data is offered (uvalid), the channel loads the word, asserts uready for one clock cycle, and sends the start bit (low). It then sends the 64 data bits, least significant first, then the stop bit (high). It then waits DELAY further bit periods, with the line high, before it may load the next word. With DELAY = 0, frames follow each other with no gap: 66 bit periods per word. DELAY throttles the transmission rate, to avoid overwhelming the ASICs.

Figure 4 shows the frames of the two packets of Fig. 3. UART 2, selected only in packet 1, sends one

frame; UARTs 0 and 1 send their second word directly after the stop bit and the delay of the first.

4.4 TX registers

Table 5: Registers of the TX unit (scope `00`). Per UART registers are at channel 0–39 (`0x00`–`0x27`); UART_CONFIG can also be written at the broadcast channel (`0x3B`). Common registers are at channel `0x3F`. R is the register byte of the address.

Register	R	Channel	Access	Contents
UART_STATUS	`0x00`	UART	RO	channel status (Table 6)
UART_CONFIG	`0x04`	UART	RW	PHASE and DELAY (Table 6); default `0x00000006`
UART_STARTS	`0x20`	UART	RO	counter: frames started (start bits sent)
UART_BEATS	`0x24`	UART	RO	counter: words taken from the TX buffer (uvalid and uready)
LOOK_SELECT	`0xA0`	common	RW	the UART shown by LOOK_LOWER and LOOK_UPPER
LOOK_LOWER	`0xA4`	common	RO	that UART's word in the output buffer, bits 31–0
LOOK_UPPER	`0xA8`	common	RO	that UART's word in the output buffer, bits 63–32
BUFFER_STATUS	`0xB0`	common	RO	TX buffer status (Table 6)
ZERO_CNTS	`0xF8`	common	WO	clear all TX counters (command: any value)

Table 6: Bit fields of the TX registers.

Register	Field	Bits	Description
UART_CONFIG	PHASE	7–0	clock cycles from the BAUD strobe to the bit edges
	DELAY	31–16	bit periods of idle line between frames
UART_STATUS	START	0	a start bit was sent (one clock cycle)
	TX	1	the POSI line
	VALID	2	uvalid (a word is waiting for this UART)
	READY	3	uready (the word is taken; one clock cycle)
	STATE	5–4	0 idle, 1 shifting, 2 stop bit, 3 delay
	BIT	6	the bit-edge strobe (BAUD + PHASE)
BUFFER_STATUS	TREADY	0	stream tready
	TVALID	1	stream tvalid
	RELEASE	4	the stream reader is released (one clock cycle)
	PACKET	5	the stream reader holds a complete packet
	STATE	9–8	0 idle, 1 start of transmission, 2 transmitting

The registers of the TX unit are listed in Table 5, and their bit fields in Table 6. The configuration of each UART (UART_CONFIG) holds PHASE and DELAY; the bit period itself is set for all UARTs by the Timing unit (CONFIG_BAUD). The counters count, for each UART, the frames started (UART_STARTS) and the words taken from the TX buffer (UART_BEATS); the two are equal in normal operation. The look registers show the word waiting in, or last loaded from, the output buffer for one UART: they check the path from DMA to the TX buffer, not the UART transmitter itself, which can be checked by the RX loopback (Section 5.1). The status registers are snapshots, mainly useful for debugging.

5 RX unit

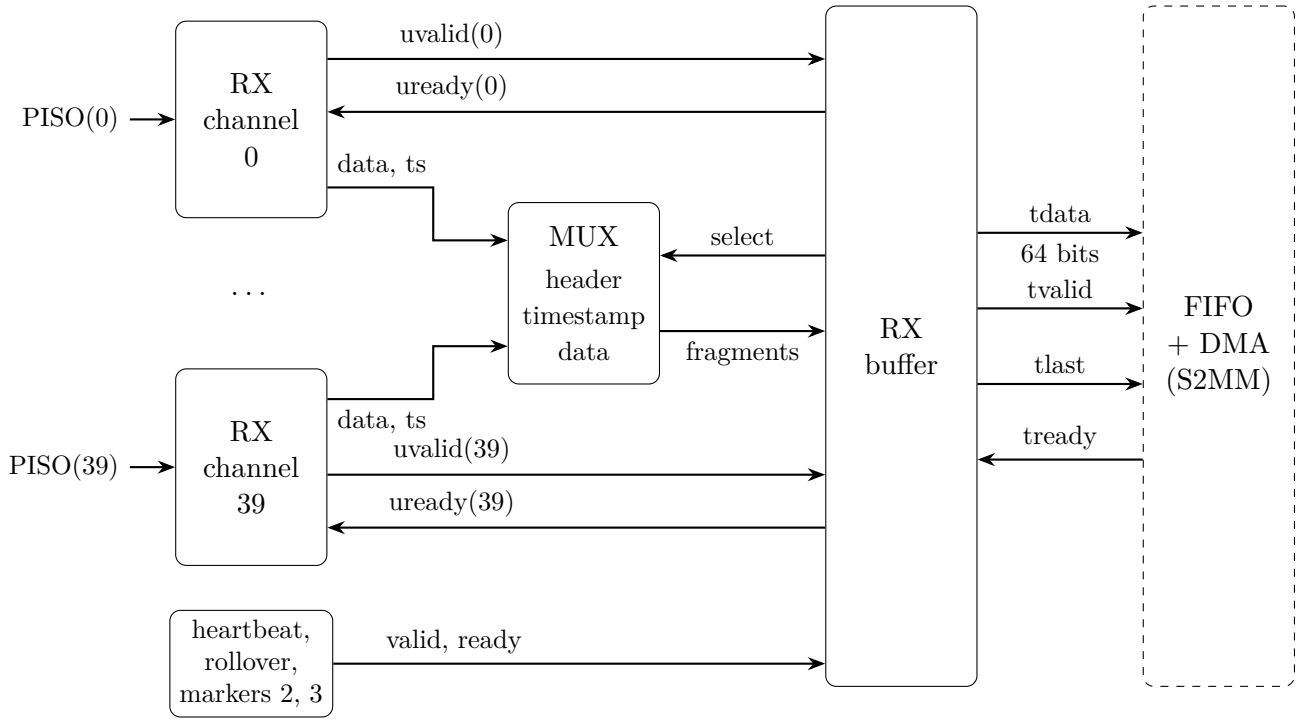


Figure 5: Data path of the RX unit. Each RX channel presents its word (data and timestamp) with a valid bit. The RX buffer selects one channel at a time; the multiplexer presents that channel's three 64-bit fragments (header, timestamp, data), which the buffer writes to the stream. The heartbeat, rollover and marker sources are four further channels (their timestamps also pass the multiplexer).

The RX unit receives the UART words from the tiles and streams them to the PS. Its data path is shown in Fig. 5. Forty RX channels receive the PISO lines independently. Each completed word is held, with its timestamp, until the RX buffer takes it. The RX buffer visits the channels in turn and writes each waiting word to the AXI stream, as a 192-bit stream word of three 64-bit fragments: a header, the timestamp, and the data. It closes the stream into DMA packets (TLAST). An AXI stream FIFO crosses into the PS clock domain, and the DMA engine writes the packets to memory.

5.1 RX channel

Each RX channel is a UART receiver with a 64-bit shift register. The input is chosen by the INPUT field of the channel's configuration: the PISO line from the tile (normal operation), the channel's own TX output (POSI, a loopback through the PL only), the RX test pattern (Section 5.4), or none (the line held high). The selected input is asynchronous to the PL clock. It passes a two-stage synchronizer, and then a three-sample history, which gives a majority vote of the last three samples, a flag when they disagree (dissent), and a glitch-free falling-edge detection (two low samples after a high one).

The receiver samples the line once per bit period. In the normal mode (MODE 1), the sample point follows the start bit: while the channel is idle, the falling edge of a start bit is located within the bit period (the number of clock cycles since the last BAUD strobe, the *edge phase*), and the sample point is set PHASE clock cycles later, modulo PERIOD. PERIOD must equal the bit period set in the Timing unit (CONFIG_BAUD; 10 by default). The default PHASE of 4 puts the sample near the middle of the bit, allowing for the delay of the synchronizer. In the fixed mode (MODE 2), the sample point is PHASE clock cycles after each BAUD strobe, whatever the input. With MODE 0 the channel never samples, so

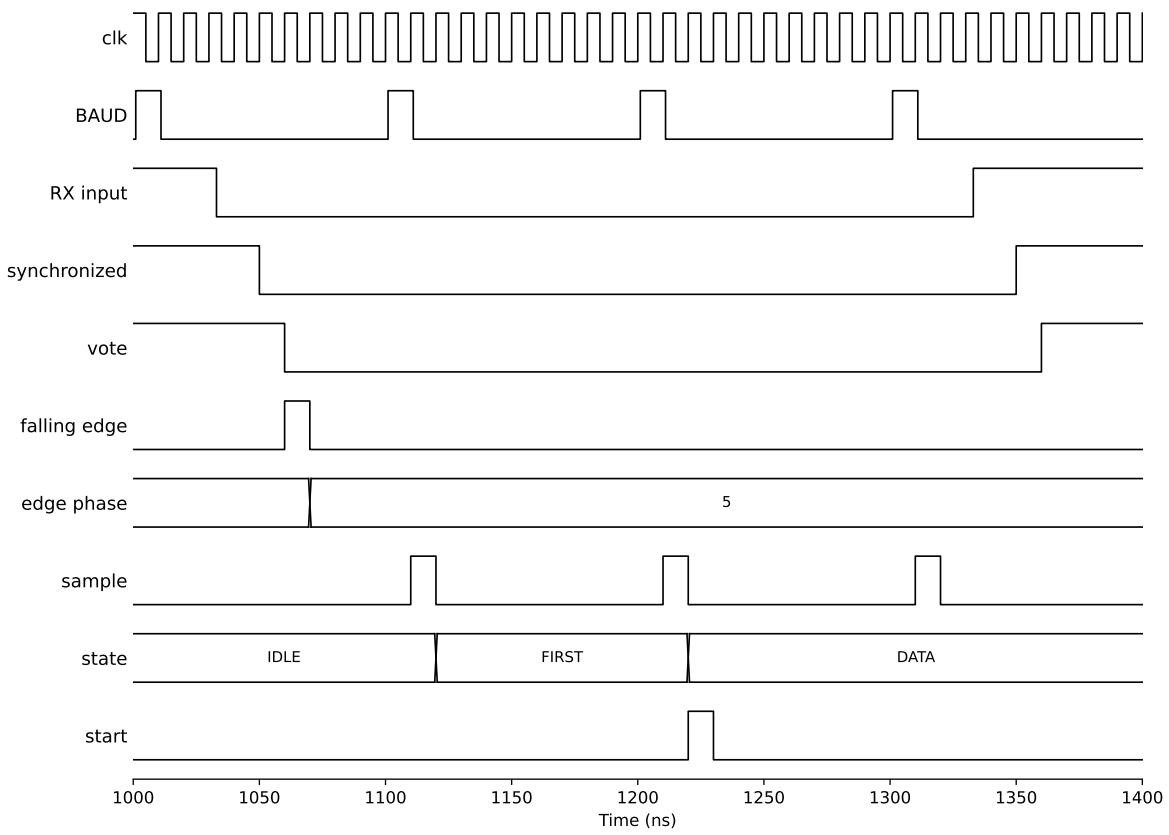


Figure 6: An RX channel finding the start bit (GHDL simulation of `rx_chan`; default configuration `0x000A1104`: PERIOD 10, MODE 1, INPUT 1, PHASE 4; BAUD every 10 clock cycles). The input falls at 1033 ns, asynchronously to the clock. After the two-stage synchronizer and the three-sample history, the falling edge is detected five clock cycles after the BAUD strobe (edge phase 5). The sample point is set PHASE = 4 clock cycles after the edge phase, in every bit period: the first sample sees the start bit (state FIRST), the next one data bit 0, where the start flag is raised, and every following sample one more data bit.

it receives nothing.

At each sample point the majority vote is taken. A low vote while idle is a start bit: the following 64 samples are the data bits, least significant first, and the next sample is the stop bit (Fig. 6). A start bit is accepted only if its three samples agree, unless flag bit 14 of the configuration is set. A sample at which the three samples disagree is counted as noise.

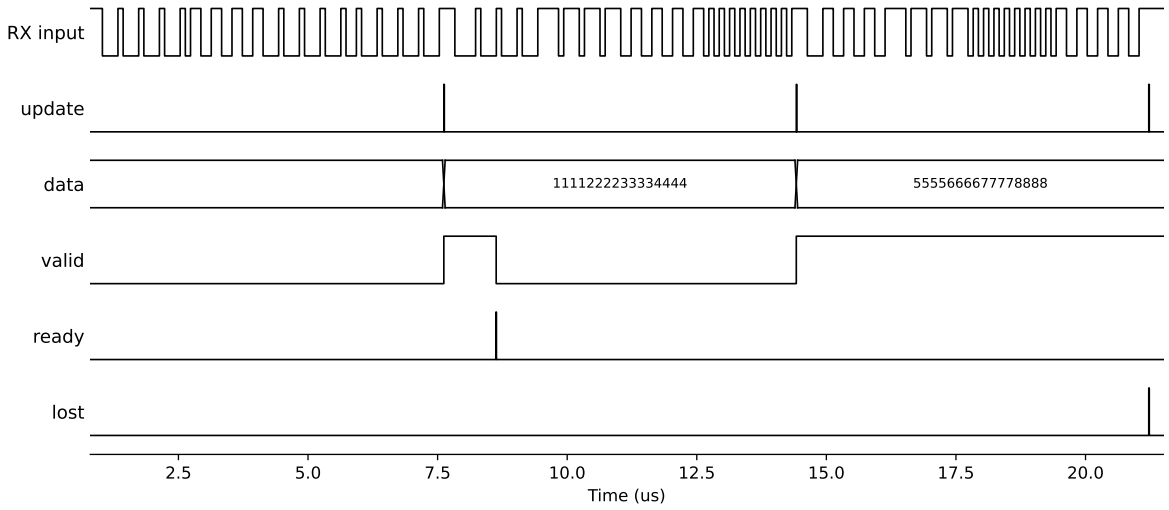


Figure 7: An RX channel receiving three frames back to back (same simulation as Fig. 6). The first word is taken by the RX buffer (ready) soon after it is valid; the second is held, valid, since it is not taken; the third arrives while the second is still valid, and is lost.

A high stop bit completes the frame (update): the word is copied to the channel's output, together with the timestamp of the Timing unit at that moment, and the channel asserts valid. Valid is cleared when the RX buffer takes the word (ready). The output holds one word only: if a further frame completes while valid is still set, the new word is discarded and counted as lost (Fig. 7). A low stop bit is a framing error: the word is discarded and counted. In normal operation the RX buffer takes each word within one sweep of its channels (135 clock cycles, Section 5.3), far less than the 660 clock cycles of a frame, so no word is lost.

5.2 Stream words

Each word in the stream is 192 bits, sent as three 64-bit fragments (Table 7). The header identifies the word. Its word type is one byte of the WORD_TYPE_LUT register, chosen by the two least significant bits of the data (for LArPix, the packet type); by default all four bytes are `0x44`, the character 'D'. The PACMAN identifier is set by the PACMAN register, and the channel identifier of each UART by its UART_CHAN register, so that the PS can label the data without further bookkeeping.

Besides the 40 UARTs, the RX buffer serves four extra channels (Table 8), each of which adds a word with a timestamp to the stream:

- **heartbeat** (channel 40): a word every HEARTBEAT_CONFIG clock cycles (default `0x3B9ACA00`, 10^9 clock cycles: every 10 s). A heartbeat that is not yet taken delays the next one; none is lost.
- **rollover** (channel 41): a word each time the least significant 32 bits of the timestamp equal ROLLOVER_CONFIG. The default, 1, marks each wrap of the 32-bit timestamp: once every 2^{32} periods of the ASIC clock, about 7.2 minutes at 10 MHz.
- **markers 2 and 3** (channels 42 and 43): a word for each pulse of marker 2 or 3 of the Timing unit (Section 6), for instance for a LEMO trigger input.

Table 7: An RX stream word: three 64-bit fragments. For the extra channels (40–43) the header is the HEADER_A–D register with the PACMAN identifier in bits 15–8, and the data fragment is zero.

Fragment	Bits	Contents
header	7–0	word type: WORD_TYPE_LUT byte chosen by the two least significant data bits
	15–8	PACMAN identifier (PACMAN register, bits 7–0)
	31–16	channel identifier (UART_CHAN register of the UART; default UART number + 1)
	63–32	zero
timestamp	63–0	timestamp at the completion of the word (Timing unit)
data	63–0	the received UART word

Table 8: The 44 channels of the RX buffer, visited in this order. The extra channels are each enabled by one bit of BUFFER_ENABLES (all off by default).

Turn	Source	Header (default)	Enable
0–39	UART 0–39	channel id, PACMAN id, word type	always
40	heartbeat	HEADER_A (`0x00480053`, type 'S')	BUFFER_ENABLES bit 0
41	rollover	HEADER_B (`0x00530053`, type 'S')	BUFFER_ENABLES bit 1
42	marker 2	HEADER_C (`0x00000054`, type 'T')	BUFFER_ENABLES bit 2
43	marker 3	HEADER_D (`0x00000000`)	BUFFER_ENABLES bit 3
trailer	end of packet	EOP_HEADER (`0x0000004C`, type 'L'), then the word count, then zero	

The header of an extra-channel word is the register HEADER_A to HEADER_D, with bits 15–8 replaced by the PACMAN identifier, and its data fragment is zero. The packet trailer (below) is a word too: its header is EOP_HEADER, also with the PACMAN identifier in bits 15–8, its second fragment the number of words in the packet (not counting the trailer), and its third fragment zero.

5.3 RX buffer

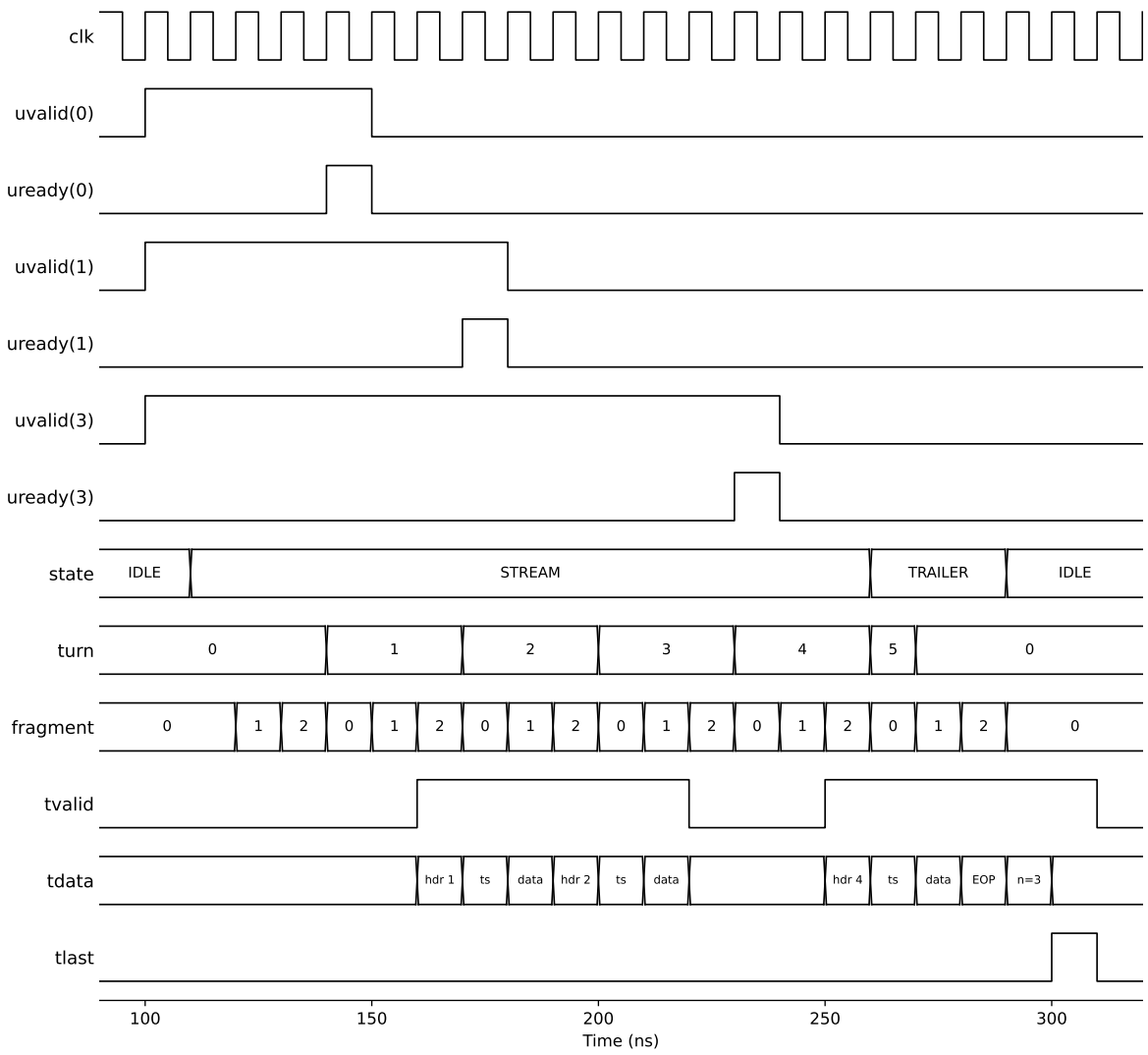


Figure 8: The RX buffer streaming three words and the trailer (GHDL simulation of `rx_buffer`; BUFFER_CONFIG `0x00030000`: at most 3 words per packet, no timeout). UARTs 0, 1 and 3 hold words. On tdata, "hdr n" is a header with channel identifier n (UART $n-1$), "ts" the timestamp, "data" the data; the trailer is EOP, the count $n = 3$, and a zero fragment with tlast. Turn 2 (UART 2, no data) leaves a gap in the stream. The third word fills the packet, so the trailer follows at once.

The RX buffer is a state machine with four states: IDLE (no data), STREAM, WAIT (all waiting words sent, packet still open) and TRAILER. When any channel has a word waiting, the buffer leaves IDLE and enters STREAM, starting at turn 0 (or, after a packet closed by the maximum, at the turn it cut short; see below). Each turn lasts three clock cycles, one per fragment, and visits one channel, in the order of Table 8. In the first clock cycle of a turn the buffer selects the channel; the multiplexers present

that channel's header, timestamp and data; in the third clock cycle, if the channel has a word waiting, the buffer copies the three fragments and asserts that channel's ready for one clock cycle, which clears its valid. The fragments are written to the stream during the following turn. A channel with no word costs its three clock cycles, but writes nothing (Fig. 8). A full sweep of the 44 channels takes 135 clock cycles. After the sweep, the buffer goes back to WAIT, and from WAIT starts a new sweep at turn 0 as soon as a word is waiting. Each sweep therefore starts at UART 0, which keeps the words of a packet in UART order when the tiles send synchronously.

The stream is closed into DMA packets by the trailer. BUFFER_CONFIG sets two conditions, each disabled by the value zero: a maximum number of words (bits 31–16) and a timeout (bits 15–0, in units of 256 clock cycles, 2.56 μs). When the packet holds the maximum number of words, the buffer sends the trailer at the end of the current turn (Fig. 8), and the next packet resumes the sweep at the channel of that turn, which was not served; so channels that are busy all the time cannot starve the ones after them. When the buffer has no word waiting (IDLE), the resume point returns to turn 0. The timeout runs from the first word of the packet, and is checked only in WAIT, between sweeps: once it has expired, the trailer follows the current sweep. The default, `0x00000001` (no maximum, timeout 256 clock cycles), therefore closes the packet at the end of the first sweep that ends 256 clock cycles or more after its first word, at most 256 + 136 = 392 clock cycles after it (a sweep and the WAIT clock cycle): all the words of one sweep travel in one packet, a steady flow of data is cut into packets of about 256 clock cycles, and since a UART word takes 66 bit periods (396 clock cycles at BAUD 6), no channel has two words in one packet at BAUD 6 or slower. The trailer's last fragment carries TLAST; the buffer then returns to IDLE.

The buffer never drops data itself. If the FIFO is full, the stream stalls (tready low), the buffer holds its current fragments, and the channels keep their words; only a channel that receives a further word before its word is taken loses data, as counted by UART_LOST. The FIFO fill level is available as FIFO_CNT, and its maximum since the last clear as FIFO_MAX.

5.4 Test pattern

The RX unit contains a test pattern generator, which produces a UART stream that any RX channel can receive in place of its PISO input (INPUT = 4). When on (PATTERN_CONFIG bits 11–8 = 1), it repeatedly sends the 64-bit word PATTERN_UPPER:PATTERN_LOWER as a UART frame (start bit, 64 bits least significant first, stop bit), followed by PATTERN_DELAY bit periods of idle line. It uses the BAUD strobe for its bit timing, with the PHASE in PATTERN_CONFIG bits 7–0, like a TX channel. All channels set to INPUT = 4 receive the same frames at the same time, which exercises the whole RX unit, buffer and DMA, at full rate and without any tile. The frames sent are counted (PATTERN_STARTS, PATTERN_STOPS).

5.5 RX registers

The registers of the RX unit are listed in Tables 9 and 10, and their bit fields in Table 11. The counters describe each UART completely: in normal operation STARTS, UPDATES and BEATS are equal, and LOST, FRAME_ERR and NOISE are zero. The look registers show the last word received by one UART (lost words and words with a framing error excepted). As in the TX unit, the status registers are snapshots, mainly useful for debugging.

6 Timing unit

The Timing unit (ATC, for ASIC timing and control, in the firmware sources) produces everything that sets the time of the system: the ASIC clock, the bit timing of the UARTs, the timestamp, and two control signals to the ASICs on each tile, G and H. Its structure is shown in Fig. 9.

Table 9: Registers of the RX unit for each UART (scope 01, channel 0–39, 0x00–0x27). UART_CONFIG can also be written at the broadcast channel (0x3B).

Register	R	Access	Contents
UART_STATUS	0x00	RO	channel status (Table 11)
UART_CONFIG	0x04	RW	Table 11; default 0x000A1104
UART_CHAN	0x08	RW	channel identifier in the header (bits 15–0); default UART + 1
UART_STARTS	0x20	RO	counter: start bits received
UART_BEATS	0x24	RO	counter: words taken by the RX buffer
UART_UPDATES	0x28	RO	counter: frames completed (good stop bit)
UART_LOST	0x2C	RO	counter: words lost (previous word not yet taken)
UART_FRAME_ERR	0x30	RO	counter: framing errors (low stop bit)
UART_NOISE	0x34	RO	counter: sample points with dissenting samples

Table 10: Registers of the RX unit in common (scope 01, channel 0x3F: addresses 0x7F00 + R).

Register	R	Access	Contents
PATTERN_LOWER	0x80	RW	test pattern word, bits 31–0
PATTERN_UPPER	0x84	RW	test pattern word, bits 63–32
PATTERN_DELAY	0x88	RW	idle bit periods between test frames
PATTERN_CONFIG	0x8C	RW	bits 7–0 PHASE, bits 11–8 MODE (1: on)
PATTERN_STATUS	0x90	RO	bit 0 start, bit 1 stop, bit 8 the pattern line
PATTERN_STARTS	0x94	RO	counter: test frames started
PATTERN_STOPS	0x98	RO	counter: test frames completed
LOOK_SELECT	0xA0	RW	the UART shown by LOOK_LOWER and LOOK_UPPER
LOOK_LOWER	0xA4	RO	that UART's last word, bits 31–0
LOOK_UPPER	0xA8	RO	that UART's last word, bits 63–32
BUFFER_STATUS	0xB0	RO	RX buffer status (Table 11)
BUFFER_CONFIG	0xB4	RW	bits 31–16 maximum words, bits 15–0 timeout; default 0x00000001
BUFFER_ENABLES	0xB8	RW	extra channels: bit 0 heartbeat, 1 rollover, 2 marker 2, 3 marker 3
PACMAN	0xBC	RW	PACMAN identifier in the header (bits 7–0)
HEARTBEAT_CONFIG	0xC0	RW	clock cycles between heartbeats; default 0x3B9ACA00
ROLLOVER_CONFIG	0xC4	RW	timestamp (bits 31–0) of the rollover word; default 1
WORD_TYPE_LUT	0xC8	RW	word type for data bits 1–0 = 0, 1, 2, 3 (bytes 0–3); default 0x44444444
HEADER_A–D	0xD0–0xDC	RW	headers of channels 40–43 (Table 8); bits 15–8 are replaced by the PACMAN identifier in the stream
EOP_HEADER	0xE0	RW	header of the trailer, bits 15–8 replaced likewise; default 0x0000004C
FIFO_CNT	0xF0	RO	words (64 bits) in the RX FIFO
FIFO_MAX	0xF4	RO	maximum of FIFO_CNT since the last clear
ZERO_CNTS	0xF8	WO	clear all RX counters and FIFO_MAX (command: any value)

Table 11: Bit fields of the RX registers.

Register	Field	Bits	Description
UART_CONFIG	PHASE	7–0	MODE 1: clock cycles from the start-bit edge to the sample point; MODE 2: from the BAUD strobe
	INPUT	11–8	one-hot: 0 none (line high), 1 PISO, 2 loopback (POSI), 4 test pattern, 8 emulator (held high in this version)
	MODE	13–12	0 off, 1 sample point from the start bit, 2 fixed sample point
	FLAGS	15–14	bit 14: accept a start bit with dissenting samples
	PERIOD	23–16	the bit period in clock cycles (equal to the Timing unit's CONFIG_BAUD)
UART_STATUS	STATE	1–0	0 idle, 1 start bit, 2 data, 3 stop bit
	HANDSHAKE	3–2	ready (bit 3) and valid (bit 2) with the RX buffer
	EVENTS	7–4	frame error, lost, update, start (bits 7–4; one clock cycle each), as counted
	SAMPLING	11–8	dissent, sample point, BAUD, noise (bits 11–8)
	TAPS	15–12	majority vote (bit 15) and the three-sample history (bits 14–12)
	INPUTS	19–16	emulator, test pattern, loopback, PISO (bits 19–16)
	EDGE	31–24	edge phase: position of the last start-bit edge in the bit period
BUFFER_STATUS	STATE	2–0	0 idle, 1 wait, 2 stream, 3 trailer
	STREAM	7–3	stream writer: last, write, busy, tready, tvalid (bits 7–3)
	TURN	13–8	current turn (channel)
	FRAGMENT	15–14	current fragment

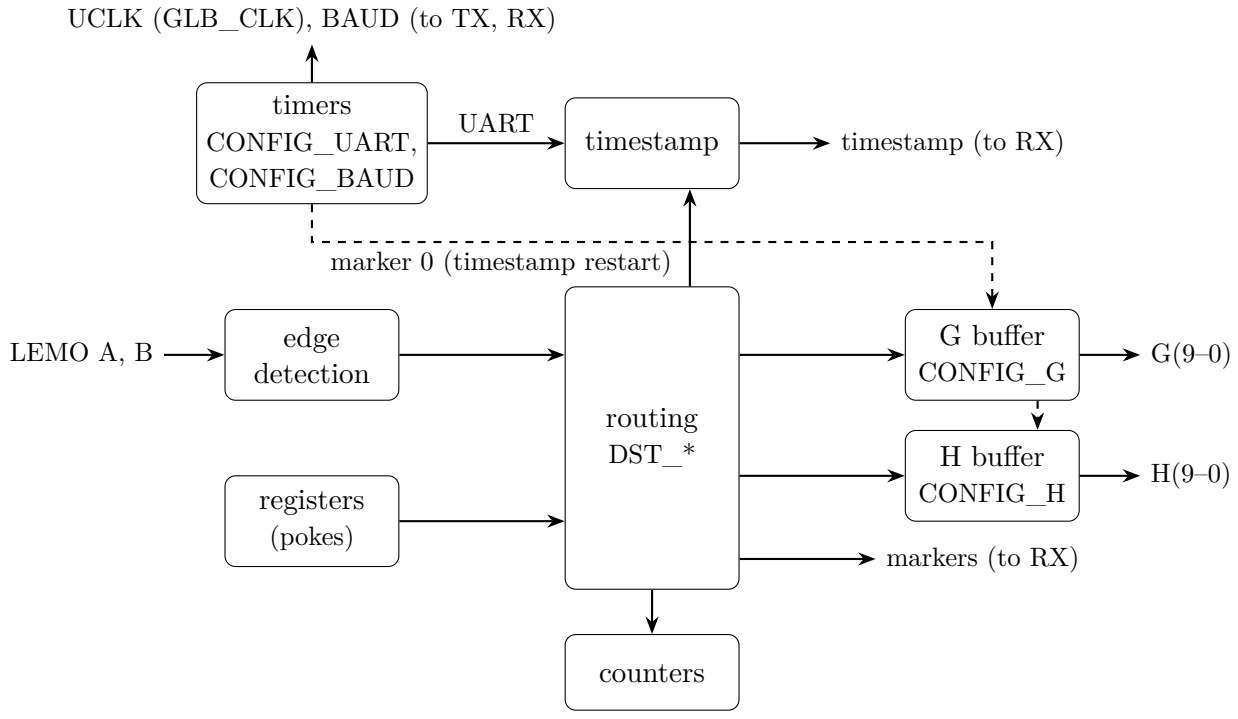


Figure 9: The Timing unit. The timers set the ASIC clock (UCLK), the UART strobe and the BAUD strobe. The routing sends each stimulus (LEMO inputs, register pokes) to the G and H outputs of selected tiles, and to the markers; the G and H buffers align the outputs to the ASIC clock (dashed). G and H are abstract control signals: their meaning on each board is in Table 12.

G and H are abstract on purpose: the firmware does not name them after a function, because the PACMAN schematics and the ASIC disagree on what the lines do. On hw1v5 the line the schematics call SYNC/RST is the v3 ASIC's reset, and the schematics' TRIG line is the ASIC's SYNC (its trigger function is hardly used). The meaning of G and H is set by the board's block design (Table 12).

Table 12: G and H on each board.

Board	G	H
hw1v5	port SYNC_O_0; schematic SYNC/RST; v3 ASIC RST	port TRIG_O_0; schematic TRIG; v3 ASIC SYNC (and TRIG)
hw1v4	legacy_1v4 block, input SYNCN_I	legacy_1v4 block, input TRIG_I

6.1 Clocks and timestamp

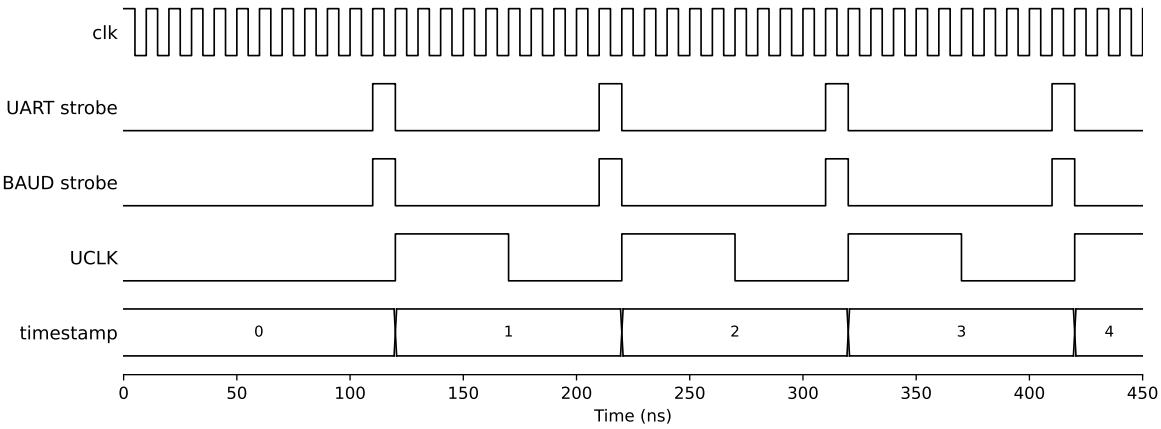


Figure 10: The strobes, the ASIC clock and the timestamp after reset (GHDL simulation of `atc_unit`; default periods of 10 clock cycles). UCLK rises on the clock cycle after each UART strobe and falls half a period later; the timestamp counts the UART strobes.

Two independent timers count the 100 MHz clock. The UART timer produces a one-clock strobe every CONFIG_UART clock cycles, and the BAUD timer a strobe every CONFIG_BAUD clock cycles (both 10 by default: 10 MHz). The UART strobe makes the ASIC clock, UCLK, which is sent to all tiles (GLB_CLK): it rises after each UART strobe and falls after half the period, a 50% duty cycle at the default period. The BAUD strobe sets the bit period of all TX and RX channels. The two are independent, though normally equal.

The timestamp is a 64-bit count of UART strobes, that is, of periods of the ASIC clock (Fig. 10). It is attached by the RX unit to every word received, and its least significant 32 bits can be read in the TIMESTAMP register. Marker 0 restarts the timestamp (Fig. 11). Since one stimulus can be routed both to marker 0 and to G or H, the timestamp can be restarted together with the ASICs (on hw1v5 with the v3 ASIC: marker 0 and H).

6.2 Stimuli, routing and outputs

Each signal that the Timing unit can send to the tiles starts as a *stimulus*, a one-clock pulse. There are eight stimuli: the two front-panel LEMO inputs (A and B), four *pokes* (A to D), made by writing to the POKE_A to POKE_D registers, and two internal sources (LOGIC A and B), which are disabled in

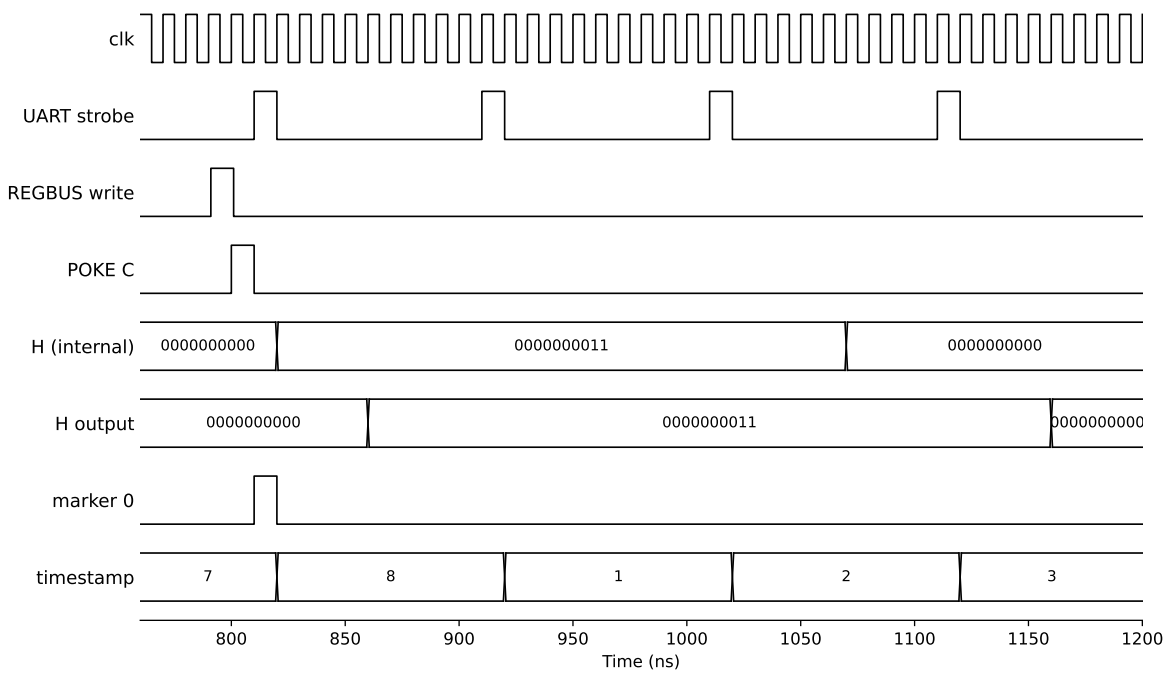


Figure 11: A poke routed to H (GHDL simulation of `atc_unit`). DST_POKE_C is `0x001903C6`: H output, marker 0, tiles 0–3, 25 clock cycles; CONFIG_H is 3. The write to POKE_C (`0x00000003`: tiles 0 and 1) makes a one-clock poke. The internal H signal is high for 25 clock cycles on tiles 0 and 1 only (the destination and the poke's own mask); the output is aligned to the UART strobe plus 3 clock cycles, and so lasts three UART periods. Marker 0 restarts the timestamp.

this version. Each stimulus has a destination register (DST_LEMO_A to DST_LOGIC_B, Table 13), which says where it goes and for how long: to the G outputs, the H outputs, or both, on the tiles of its tile mask, for a duration of a set number of clock cycles; and to any of the four markers, as a one-clock pulse. A stimulus with a destination of zero goes nowhere, which is the default.

Table 13: Bit fields of the destination registers DST_LEMO_A to DST_LOGIC_B.

Field	Bits	Description
G	0	send to the G outputs
H	1	send to the H outputs
MARKERS	5–2	send to markers 0–3 (one clock cycle)
TILES	15–6	tile mask for G and H: bit 6 for tile 0, ..., bit 15 for tile 9
DURATION	31–16	length of the G and H pulse, in clock cycles

A poke carries a further tile mask of its own, in bits 9–0 of the value written to POKE_A to POKE_D: the pulse reaches a tile only if both the destination and the poke select it. The PS can therefore pulse any set of tiles with a single write, without changing the destination. A new stimulus restarts the duration of its own pulse; pulses of different stimuli on the same output are combined.

The G and H signals of each tile pass an output buffer before they leave the PL. The buffer samples its input once per UART period, at a point set by PHASE (CONFIG_G and CONFIG_H bits 7–0, in clock cycles after the UART strobe), and holds its output until the next sample, so that G and H change at a fixed point of the ASIC clock and last a whole number of ASIC clock periods. A pulse shorter than one UART period may fall between two samples and be missed: the DURATION of a destination with G or H should be at least the UART period (10 clock cycles by default). The buffer can also invert the signal of each tile (bits 17–8, one per tile); an inverted output is high when idle. Figure 11 shows a poke on its way through the unit.

The LEMO inputs are asynchronous. Each passes a two-stage synchronizer, and a rising edge is detected when the synchronized input is high on two consecutive clock cycles after being low; no further edge is detected during the next few clock cycles. A pulse must therefore be seen high on two clock cycles: one shorter than 10 ns is always ignored, and one shorter than 20 ns may be (Fig. 12). CONFIG_INPUT bits 0 and 1 invert LEMO A and B, so that a falling edge is detected instead.

The markers connect the Timing unit to the data. Marker 0 restarts the timestamp. Markers 2 and 3 are channels 42 and 43 of the RX buffer: when enabled there, each marker pulse adds a word with its timestamp to the stream. A LEMO trigger input routed to marker 2, for example, records the time of every external trigger along with the data. Marker 1 is only counted.

6.3 Counters

The Timing unit counts its signals: the stimuli (LEMO and pokes), the G and H signals of each tile (before the output buffers), and the markers. Each counter holds two 16-bit numbers, both saturating: the number of rising edges, and the *integral*, the number of clock cycles during which the signal was high (counted only after its first rising edge since the counters were cleared). The counters are controlled by commands written to COUNT_REQ (Table 14). A READ command latches one counter into the COUNT register, as integral (bits 31–16) and edges (bits 15–0), where it can then be read. The counters run from reset; STOP, START and CLEAR act on all of them together.

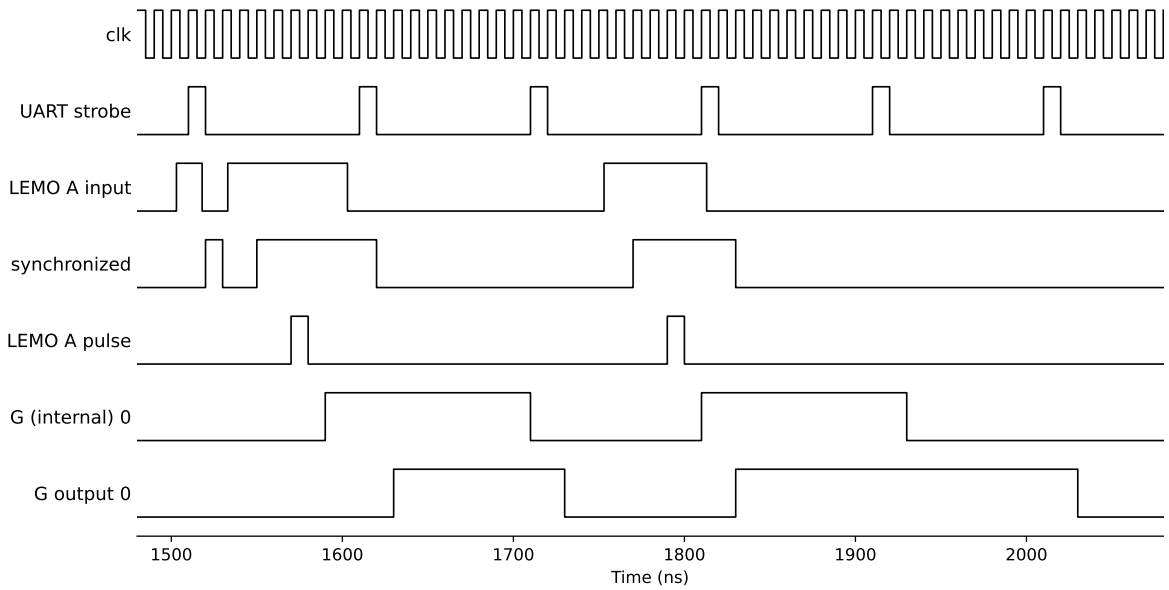


Figure 12: A LEMO input routed to G (GHDL simulation of `atc_unit`; DST_LEMO_A `0x000CFFC1`: G output, all tiles, 12 clock cycles; CONFIG_G 0). The first high, 15 ns long, is seen by the synchronizer for one clock cycle only, which is not an edge. Each rising edge after it makes one pulse, which is routed to G for 12 clock cycles and aligned to the UART strobe.

Table 14: Commands of the counters, written to COUNT_REQ (bits 7–0).

Command	Code (bits 7–4)	Description (bits 3–0: the channel C)
NO OP	0	nothing
CLEAR	1	clear all counters
START	2	start all counters (the default after reset)
STOP	3	stop all counters
READ M	4	latch the counter of marker C (0–3) into COUNT
READ G	5	latch the counter of the G signal of tile C (0–9)
READ H	6	latch the counter of the H signal of tile C (0–9)
READ I	7	latch the counter of stimulus C: 0 LEMO A, 1 LEMO B, 2–5 POKE A–D

Table 15: Registers of the Timing unit (scope `1110`).

Register	Address	Access	Contents
STATUS	`0xE000`	RO	zero (no status bits defined)
TIMESTAMP	`0xE004`	RO	timestamp, bits 31–0
POKE_A–D	`0xE0A0`, `0xE0B0`, `0xE0C0`, `0xE0D0`	WO	poke (command); bits 9–0: the poke's tile mask
CONFIG_UART	`0xE100`	RW	UART (and UCLK) period in clock cycles; default 10
CONFIG_BAUD	`0xE104`	RW	bit period of the TX and RX channels in clock cycles; default 10
CONFIG_INPUT	`0xE108`	RW	bit 0: invert LEMO A, bit 1: invert LEMO B
CONFIG_G	`0xE110`	RW	G output buffer: bits 7–0 PHASE, bits 17–8 invert (per tile)
CONFIG_H	`0xE114`	RW	H output buffer: bits 7–0 PHASE, bits 17–8 invert (per tile)
DST_LEMO_A, _B	`0xE120`, `0xE124`	RW	destination of LEMO A, B (Table 13)
DST_POKE_A–D	`0xE128`–`0xE134`	RW	destination of POKE A–D
DST_LOGIC_A, _B	`0xE138`, `0xE13C`	RW	destination of LOGIC A, B (disabled sources)
COUNT_REQ	`0xE200`	WO	counter command (Table 14)
COUNT	`0xE204`	RO	latched counter: bits 31–16 integral, bits 15–0 edges

6.4 Timing registers

The registers of the Timing unit are listed in Table 15. All configuration registers are zero after reset, except the two periods. The Timing unit answers reads and writes on the next clock cycle, like the Global unit.